

29/7/2026

Activity

(stack, queue and linked list)

1) $105 + 606 / * 8 -$

Answer:

$$10 + 5 = 15$$

$$60 \div 6 = 10$$

$$15 * 10 = 150$$

$$150 - 8 = 142$$

2) Answer:

$$\text{Stack} = 54 \rightarrow 54, 52 \rightarrow \text{pop}(52) \rightarrow 54, 55 \rightarrow 54, 55, 62 \rightarrow S$$

$$S = 62.$$

$$\text{Queue} = 21 \rightarrow 21, 24 \rightarrow \text{dequeue}(21) \rightarrow 24, 28 \rightarrow 24, 28, 32 \rightarrow$$

$$Q = 24$$

$$S + Q = 62 + 24$$

$$= 86 //$$

3) Answer:

After the first $\times$, the top 2 stack elements are 2, 16, 1, 6 //

4) Answer:

$$a b c * d e f \wedge \wedge -$$

5) Stack operation:

$$\text{Popped values: } 20 \rightarrow 26 \rightarrow 10 \rightarrow 10 \rightarrow 20.$$

6) Reverse queue:

move elements between Q_1 & Q_2 until Q_2 becomes 4, 3, 2, 1.

12 enqueue operations

7) Reverse stack:

Statement II is false //